

Large-Sample Theory for Likelihood Ratio Tests

The tests in Chapters 12 and 13 have strong optimality properties but require conditions on the densities for the data and the form of the hypotheses that are rather special and can fail for many natural models. By contrast, the generalized likelihood ratio test introduced in this chapter requires little structure, but it does not have exact optimality properties. Use of this test is justified by large-sample theory. In Section 17.2 we derive approximations for its level and power. Wald tests and score tests are popular alternatives to generalized likelihood ratio tests with similar asymptotic performance. They are discussed briefly in Section 17.4.

17.1 Generalized Likelihood Ratio Tests

Let the data $X_1, \dots, X_n$ be i.i.d. with common density f_θ for $\theta \in \Omega$. The likelihood function is

$$L(\theta) = L(\theta|X_1, \dots, X_n) = \prod_{i=1}^n f_\theta(X_i).$$

The (generalized) likelihood ratio statistic for testing $H_0 : \theta \in \Omega_0$ versus $H_1 : \theta \in \Omega_1$ is defined as

$$\lambda = \lambda(X_1, \dots, X_n) = \frac{\sup_{\Omega_1} L(\theta)}{\sup_{\Omega_0} L(\theta)}.$$

The likelihood ratio test rejects H_0 if $\lambda > k$. When H_0 and H_1 are both simple hypotheses, this test is the optimal test described in the Neyman–Pearson lemma.

Typical situations where likelihood ratio tests are used have Ω_0 a smooth manifold of smaller dimension than $\Omega = \Omega_0 \cup \Omega_1$. In this case, if $L(\theta)$ is continuous, λ can be computed as

$$\lambda = \lambda(X_1, \dots, X_n) = \frac{\sup_{\Omega} L(\theta)}{\sup_{\Omega_0} L(\theta)}.$$

Furthermore, if these supremum are attained, then

$$\lambda = \frac{L(\hat{\theta})}{L(\tilde{\theta})}, \quad (17.1)$$

where $\hat{\theta}$ is the maximum likelihood estimate of θ under the full model, and $\tilde{\theta}$ is the maximum likelihood under H_0 , with θ varying over Ω_0 .

Example 17.1. Suppose $X_1, \dots, X_n$ are a random sample from $N(\mu, \sigma^2)$ and $\theta = (\mu, \sigma)$. The log-likelihood function is

$$l(\theta) = \log L(\theta) = -\frac{n}{2} \log(2\pi\sigma^2) - \sum_{i=1}^n \frac{(X_i - \mu)^2}{2\sigma^2}.$$

The partial derivative with respect to μ is

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu).$$

Setting this equal to zero gives

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

as the value for μ that maximizes l , regardless of the value of σ , so $\hat{\mu}$ is the maximum likelihood estimate for μ . We can find the maximum likelihood estimate $\hat{\sigma}$ of σ by maximizing $l(\hat{\mu}, \sigma)$ over $\sigma > 0$. Setting

$$\frac{\partial}{\partial \sigma} l(\hat{\mu}, \sigma) = -\frac{n}{\sigma} + \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^3}$$

equal to zero gives

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Using these values in the formula for l , after some simplification

$$L(\hat{\mu}, \hat{\sigma}) = \frac{e^{-n/2}}{(2\pi\hat{\sigma}^2)^{n/2}}. \quad (17.2)$$

Suppose we wish to test $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$. The maximum likelihood estimate for μ under the null hypothesis is $\tilde{\mu} = 0$ (of course). Setting

$$\frac{\partial}{\partial \sigma} l(0, \sigma) = -\frac{n}{\sigma} + \sum_{i=1}^n \frac{X_i^2}{\sigma^3}$$

equal to zero gives

$$\tilde{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

as the maximum likelihood estimate for σ^2 under H_0 . After some algebra,

$$L(\tilde{\mu}, \tilde{\sigma}) = \frac{e^{-n/2}}{(2\pi\tilde{\sigma}^2)^{n/2}}. \quad (17.3)$$

Using (17.2) and (17.3) in (17.1), the likelihood ratio statistic is

$$\lambda = \frac{\tilde{\sigma}^n}{\hat{\sigma}^n}.$$

Using the identity

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2,$$

we have

$$\begin{aligned} \lambda &= \left[\frac{\sum_{i=1}^n X_i^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right]^{n/2} \\ &= \left[\frac{\sum_{i=1}^n (X_i - \bar{X})^2 + n\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right]^{n/2} \\ &= \left[1 + \frac{n\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right]^{n/2} \\ &= \left[1 + \frac{T^2}{n-1} \right]^{n/2}, \end{aligned}$$

where $T = \sqrt{n}\bar{X}/S$ is the t -statistic usually used to test H_0 against H_1 . Since the function relating λ to $|T|$ is strictly increasing, the likelihood ratio test is equivalent to the usual t -test, which rejects if $|T|$ exceeds a constant.

Example 17.2. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be a sample from a bivariate normal distribution. The log-likelihood is

$$\begin{aligned} l(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho) &= -\frac{1}{2(1-\rho^2)} \left[\sum_{i=1}^n \left(\frac{X_i - \mu_x}{\sigma_x} \right)^2 + \sum_{i=1}^n \left(\frac{Y_i - \mu_y}{\sigma_y} \right)^2 \right. \\ &\quad \left. - 2\rho \sum_{i=1}^n \left(\frac{X_i - \mu_x}{\sigma_x} \right) \left(\frac{Y_i - \mu_y}{\sigma_y} \right) \right] \\ &\quad - n \log(2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}). \end{aligned}$$

We derive the likelihood ratio test of $H_0 : \rho = 0$ versus $H_1 : \rho \neq 0$. When $\rho = 0$, we have independent samples from two normal distributions, and using results from the previous example,

$$\tilde{\mu}_x = \overline{X}, \quad \tilde{\mu}_y = \overline{Y}$$

and

$$\tilde{\sigma}_x^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \overline{X})^2, \quad \tilde{\sigma}_y^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \overline{Y})^2.$$

The easiest way to find the maximum likelihood estimates for the full model is to note that the family of distributions is a five-parameter exponential family, so the canonical sufficient statistic is the maximum likelihood estimate for its mean. This gives

$$\begin{aligned} \sum_{i=1}^n X_i &= n\hat{\mu}_x, & \sum_{i=1}^n Y_i &= n\hat{\mu}_y, \\ \sum_{i=1}^n X_i^2 &= n(\hat{\mu}_x^2 + \hat{\sigma}_x^2), & \sum_{i=1}^n Y_i^2 &= n(\hat{\mu}_y^2 + \hat{\sigma}_y^2), \end{aligned}$$

and

$$\sum_{i=1}^n X_i Y_i = n(\hat{\mu}_x \hat{\mu}_y + \hat{\rho} \hat{\sigma}_x \hat{\sigma}_y).$$

Solving these equations gives $\hat{\mu}_x = \tilde{\mu}_x$, $\hat{\mu}_y = \tilde{\mu}_y$, $\hat{\sigma}_x = \tilde{\sigma}_x$, $\hat{\sigma}_y = \tilde{\sigma}_y$, and

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^n (X_i - \overline{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \overline{Y})^2}}.$$

Using (17.1),

$$\begin{aligned} \log \lambda &= \log L(\overline{X}, \overline{Y}, \hat{\sigma}_x, \hat{\sigma}_y, \hat{\rho}) - \log L(\overline{X}, \overline{Y}, \hat{\sigma}_x, \hat{\sigma}_y, 0) \\ &= -\frac{n}{2} \log(1 - \hat{\rho}^2). \end{aligned}$$

Equivalent test statistics are $|\hat{\rho}|$ or $|T|$, where

$$T = \frac{\hat{\rho} \sqrt{n-2}}{\sqrt{1 - \hat{\rho}^2}}.$$

Under H_0 , T has a t -distribution on $n - 2$ degrees of freedom. In fact, the conditional distribution of T given the Y_i is t on $n - 2$ degrees of freedom. To see this, let $Z_i = (X_i - \mu_x)/\sigma_x$, and let $V_i = Y_i - \overline{Y}$. Since $X_i - \overline{X} = \sigma_x(Z_i - \overline{Z})$, we can write

$$\hat{\rho} = \frac{\sum_{i=1}^n (Z_i - \overline{Z})V_i}{\sqrt{\sum_{i=1}^n (Z_i - \overline{Z})^2} \sqrt{\sum_{i=1}^n V_i^2}}.$$

Let $a = (1, \dots, 1)' / \sqrt{n}$ and $b = V / \|V\|$. Then $\|a\| = 1$, $\|b\| = 1$, and

$$a \cdot b = \frac{1}{\sqrt{n}\|V\|} \sum_{i=1}^n (Y_i - \bar{Y}) = 0.$$

Hence we can find an orthogonal matrix O where the first two columns are a and b . Because O is constructed from Y , Z and O are independent under H_0 . By this independence, if we define transformed variables $\check{Z} = O'Z$, then $\check{Z}|O \sim N(0, I)$, which implies that $\check{Z}_1, \dots, \check{Z}_n$ are i.i.d. standard normal. Note that $\check{Z}_1 = a \cdot Z = \sqrt{n}\bar{Z}$ and $\check{Z}_2 = b \cdot Z$. Since $\|Z\| = \|\check{Z}\|$,

$$\begin{aligned} \sum_{i=1}^n (Z_i - \bar{Z})^2 &= \sum_{i=1}^n Z_i^2 - n\bar{Z}^2 \\ &= \|Z\|^2 - \check{Z}_1^2 \\ &= \sum_{i=2}^n \check{Z}_i^2, \end{aligned}$$

and hence

$$\hat{\rho} = \frac{Z \cdot b}{\sqrt{\sum_{i=2}^n \check{Z}_i^2}} = \frac{\check{Z}_2}{\sqrt{\sum_{i=2}^n \check{Z}_i^2}}.$$

From this, $1 - \hat{\rho}^2 = \sum_{i=3}^n \check{Z}_i^2 / \sum_{i=2}^n \check{Z}_i^2$, so

$$T = \frac{\check{Z}_2}{\sqrt{\frac{1}{n-2} \sum_{i=3}^n \check{Z}_i^2}}.$$

The sum in the denominator has the chi-square distribution on $n - 2$ degrees of freedom, and the numerator and denominator are independent. Therefore this agrees with the usual definition for the t -distribution.

17.2 Asymptotic Distribution of $2 \log \lambda$

In this section we derive the asymptotic distribution of $2 \log \lambda$ when $\theta \in \Omega_0$ or θ is near Ω_0 . A rigorous treatment requires considerable attention to detail and deep mathematics, at least if one is concerned with getting the best regularity conditions. To keep the presentation here as accessible as possible, we keep the treatment somewhat informal and base it on assumptions stronger than necessary. Specifically, we assume that conditions necessary for a multivariate version of Theorem 9.14 are in force: the maximum likelihood estimators $\hat{\theta}_n$ are consistent, and the densities $f_\theta(x)$ are regular enough to allow us to define the Fisher information matrix $I(\theta)$ (positive definite for all $\theta \in \Omega$ and continuous as a function of θ) and use Taylor expansion to show that

$$\sqrt{n}(\hat{\theta}_n - \theta) = I^{-1}(\theta) \frac{1}{\sqrt{n}} \nabla l_n(\theta) + o_p(1) \Rightarrow N(0, I^{-1}(\theta)). \quad (17.4)$$

The parameter space Ω is an open subset of $\mathbb{R}^r$, and Ω_0 is a smooth submanifold of Ω with dimension $q < r$. Finally, we assume that $\hat{\theta}_n$ is consistent if $\theta \in \Omega_0$.

To use likelihood ratio tests in applications, we need to know the size, so it is natural to want an approximation for the power $\beta_n(\theta)$ of the test when $\theta \in \Omega_0$. Also, to design experiments it is useful to approximate the power at other points $\theta \notin \Omega_0$. Now for fixed $\theta \in \Omega_1$, if n is large enough any reasonable test will likely reject H_0 and the power $\beta_n(\theta)$ should tend to one as $n \rightarrow \infty$. But a theorem detailing this would not be very useful in practice. For a more interesting theory we study the power at points near Ω_0 . Specifically, we study the distribution of $2 \log \lambda_n$ along a sequence of parameter values $\theta_n = \theta_0 + \Delta/\sqrt{n}$, where $\theta_0 \in \Omega_0$ and Δ is a fixed constant, and show that

$$P_{\theta_n}[2 \log \lambda_n < t] \rightarrow F(t)$$

as $n \rightarrow \infty$, where F is the cumulative distribution function for a noncentral chi-square distribution with $r - q$ degrees of freedom. When $\Delta = 0$, $\theta_n = \theta_0 \in \Omega_0$ and this result approximates the cumulative distribution function of $2 \log \lambda_n$ under H_0 . In this case, the noncentrality parameter is zero, so the likelihood ratio test, which rejects if $2 \log \lambda$ exceeds the upper α th quantile of the chi-square distribution on $r - q$ degrees of freedom, has size approximately α . Other choices for Δ allow one to approximate the power of this test.

The assumptions for Theorems 16.10 or 16.14 are weaker than those above, so the joint distributions for $X_1, \dots, X_n$ under θ_n are contiguous to the joint distributions under θ_0 , and, by Remark 16.13, under P_{θ_n}

$$\frac{1}{\sqrt{n}} \nabla l_n(\theta_0) \Rightarrow N(I(\theta_0)\Delta, I(\theta_0)).$$

By Proposition 16.5, a sequence that is $o_p(1)$ under P_{θ_0} will also be $o_p(1)$ under P_{θ_n} . If we define

$$Z_n = \sqrt{n}(\hat{\theta}_n - \theta_0) = \sqrt{n}(\hat{\theta}_n - \theta_n) + \Delta,$$

then by (17.4), under P_{θ_n} ,

$$Z_n \Rightarrow N(\Delta, I^{-1}(\theta_0)). \quad (17.5)$$

Equivalently, $\sqrt{n}(\hat{\theta}_n - \theta_n) \Rightarrow N(0, I^{-1}(\theta_0))$, showing in some sense that the usual normal approximation for the distribution of the maximum likelihood estimator holds uniformly over contiguous parameter values, which seems natural.

The rest of the argument is based on using Taylor expansion and the normal equations to relate $2 \log \lambda_n$ to Z_n . The o_p and O_p notations for scales

of magnitude, introduced in Section 8.6, provide a convenient way to keep track of the size of errors in these expansions. Note that $Z_n = O_p(1)$ by (17.5), and so $o_p(Z_n) = o_p(1)$. The normal equations for $\hat{\theta}_n$ are simple, the gradient of l_n must vanish at $\hat{\theta}_n$; that is,

$$\nabla l_n(\hat{\theta}_n) = 0. \quad (17.6)$$

The normal equations for $\tilde{\theta}_n$ involve the local geometry of Ω_0 and are more delicate. For $\theta \in \Omega_0$, let V_θ denote the tangent space¹ at θ , and let $P(\theta)$ denote the projection matrix onto V_θ . As $x \in \Omega_0$ approaches θ , $x - \theta$ should almost lie in V_θ . Specifically,

$$x - \theta = P(\theta)(x - \theta) + o(\|x - \theta\|). \quad (17.7)$$

Also, the matrices $P(\theta)$ should vary continuously with θ . When $\tilde{\theta}_n$ lies in the interior of Ω , the directional derivatives of l_n for vectors in the tangent space at $\tilde{\theta}_n$ must vanish; otherwise we could move a little in Ω_0 and increase the likelihood. So if $\tilde{P}_n \stackrel{\text{def}}{=} P(\tilde{\theta}_n)$, then

$$\tilde{P}_n \nabla l_n(\tilde{\theta}_n) = 0. \quad (17.8)$$

Also, by continuity, because $\tilde{\theta}_n \rightarrow \theta_0$, $\tilde{P}_n \xrightarrow{p} P_0 \stackrel{\text{def}}{=} P(\theta_0)$, the projection matrix onto the tangent plane of Ω_0 at θ_0 . Using this,

$$P_0 \nabla l_n(\tilde{\theta}_n) = o_p(\nabla l_n(\tilde{\theta}_n)). \quad (17.9)$$

Since $\tilde{\theta}_n$ and θ_0 are close to each other and both lie in Ω_0 , by (17.7) their normalized difference $Y_n = \sqrt{n}(\tilde{\theta}_n - \theta_0)$ satisfies

$$Y_n = P_0 Y_n + o_p(Y_n). \quad (17.10)$$

Let $\nabla^2 l_n$ denote the Hessian matrix of second partial derivatives of l_n . By the weak law of large numbers,

$$\frac{1}{n} \nabla^2 l_n(\theta) = \frac{1}{n} \sum_{i=1}^n \nabla_\theta^2 \log f_\theta(X_i) \xrightarrow{p} -I(\theta)$$

in P_θ -probability as $n \rightarrow \infty$, since

$$E_\theta \nabla_\theta^2 \log f_\theta(X_1) = -I(\theta).$$

By contiguity, $\nabla^2 l_n(\theta_0)/n \rightarrow -I(\theta_0)$ in P_{θ_n} -probability as $n \rightarrow \infty$. Also, using Theorem 9.2, our weak law for random functions, convergence in probability also holds if the Hessian is evaluated at intermediate values approaching θ_0 . Using this observation, one-term Taylor expansions of $\nabla l_n/\sqrt{n}$ about θ_0 give

¹ See Appendix A.4 for an introduction to manifolds and tangent spaces.

$$\frac{1}{\sqrt{n}}\nabla l_n(\hat{\theta}_n) - \frac{1}{\sqrt{n}}\nabla l_n(\theta_0) = (-I(\theta_0) + o_p(1))\sqrt{n}(\hat{\theta}_n - \theta_0) \quad (17.11)$$

and

$$\frac{1}{\sqrt{n}}\nabla l_n(\tilde{\theta}_n) - \frac{1}{\sqrt{n}}\nabla l_n(\theta_0) = (-I(\theta_0) + o_p(1))\sqrt{n}(\tilde{\theta}_n - \theta_0). \quad (17.12)$$

With the definition of Z_n and (17.6), the first Taylor approximation above becomes

$$\frac{1}{\sqrt{n}}\nabla l_n(\theta_0) = I(\theta_0)Z_n + o_p(1). \quad (17.13)$$

Multiplying the second Taylor approximation by P_0 and using (17.9),

$$P_0 \frac{1}{\sqrt{n}}\nabla l_n(\theta_0) = P_0 I(\theta_0)Y_n + o_p(Y_n) + o_p(1),$$

and these last two equations give

$$P_0 I(\theta_0)Z_n = P_0 I(\theta_0)Y_n + o_p(Y_n) + o_p(1). \quad (17.14)$$

We have now obtained three key equations: (17.5), (17.10), and (17.14). We also need an equation relating $2\log \lambda_n$ to Y_n and Z_n . This follows from a two-term Taylor expansion, again equating $\nabla^2 l_n$ at intermediate values with $-nI(\theta_0) + o_p(n)$, which gives

$$\begin{aligned} 2\log \lambda_n &= 2l_n(\hat{\theta}_n) - 2l_n(\tilde{\theta}_n) \\ &= 2(\hat{\theta}_n - \theta_0)' \nabla l_n(\theta_0) - (\hat{\theta}_n - \theta_0)' (nI(\theta_0) + o_p(n))(\hat{\theta}_n - \theta_0) \\ &\quad - 2(\tilde{\theta}_n - \theta_0)' \nabla l_n(\theta_0) + (\tilde{\theta}_n - \theta_0)' (nI(\theta_0) + o_p(n))(\tilde{\theta}_n - \theta_0) \\ &= 2Z_n' \frac{1}{\sqrt{n}} \nabla l_n(\theta_0) - Z_n' (I(\theta_0) + o_p(1))Z_n \\ &\quad - 2Y_n' \frac{1}{\sqrt{n}} \nabla l_n(\theta_0) + Y_n' (I(\theta_0) + o_p(1))Y_n. \end{aligned}$$

Using (17.13)

$$\begin{aligned} 2\log \lambda_n &= 2Z_n' I(\theta_0)Z_n - Z_n' I(\theta_0)Z_n + o_p(1) \\ &\quad - 2Y_n' I(\theta_0)Z_n + Y_n' I(\theta_0)Y_n + o_p(\|Y_n\|^2) \\ &= (Z_n - Y_n)' I(\theta_0)(Z_n - Y_n) + o_p(\|Y_n\|^2) + o_p(1). \end{aligned} \quad (17.15)$$

The approximation for the P_{θ_n} distributions of $2\log \lambda_n$, mentioned above, follows eventually from (17.5), (17.10), (17.14), and (17.15). The algebra for this derivation is easier if we write the quantities involved in a convenient basis. Let $V = V_{\theta_0}$ denote the tangent space of Ω_0 at θ_0 , and let $V^\perp$ denote its orthogonal complement. Then for $v \in V$, $P_0 v = v$, and for $v \in V^\perp$, $P_0 v = 0$. Let $e_1, \dots, e_q$ be an orthonormal basis for V , and let $e_{q+1}, \dots, e_r$ be an

orthonormal basis for $V^\perp$. Because $e_1, \dots, e_r$ is an orthonormal basis for $\mathbb{R}^r$, $O = (e_1, \dots, e_r)$ is an orthogonal matrix. Also, $P_0 O = (e_1, \dots, e_q, 0, \dots, 0)$, so

$$O' P_0 O \stackrel{\text{def}}{=} \check{P} = \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix},$$

where I_q denotes the $q \times q$ identity matrix and the zeros are zero matrices with suitable dimensions. In the new basis the key variables are $\check{Y} = O' Y_n$, $\check{Z} = O' Z_n$, $\check{I} = O' I(\theta_0) O$, and $\check{\Delta} = O' \Delta$. By (17.5), $\check{Z} \Rightarrow N_r(O' \Delta, O' I^{-1}(\theta_0) O)$, and since $O' I^{-1}(\theta_0) O = \check{I}^{-1}$, this becomes

$$\check{Z} \Rightarrow N_r(\check{\Delta}, \check{I}^{-1}). \quad (17.16)$$

Premultiplying (17.10) by O' and inserting $O' O$ at useful places gives

$$O' P_0 O O' Y_n = O' Y_n + o_p(Y_n),$$

or

$$\check{P} \check{Y} = \check{Y} + o_p(\check{Y}). \quad (17.17)$$

Similarly, premultiplying (17.14) by O' gives

$$O' P_0 O O' I(\theta_0) O O' Y_n = O' P_0 O O' I(\theta_0) O O' Z_n + o_p(Y_n) + o_p(1),$$

or

$$\check{P} \check{I} \check{Y} = \check{P} \check{I} \check{Z} + o_p(\check{Y}) + o_p(1). \quad (17.18)$$

Finally, (17.15) gives

$$2 \log \lambda_n = (Z_n - Y_n) O O' I(\theta_0) O O' (Z_n - Y_n) + o_p(\|Y_n\|) + o_p(1),$$

which becomes

$$2 \log \lambda_n = (\check{Z} - \check{Y})' \check{I} (\check{Z} - \check{Y}) + o_p(\|\check{Y}\|) + o_p(1). \quad (17.19)$$

To continue we need to partition $\check{Z}$, $\check{Y}$, and $\check{I}$ as

$$\check{Z} = \begin{pmatrix} \check{Z}_1 \\ \check{Z}_2 \end{pmatrix}, \quad \check{Y} = \begin{pmatrix} \check{Y}_1 \\ \check{Y}_2 \end{pmatrix}, \quad \check{I} = \begin{pmatrix} \check{I}_{11} & \check{I}_{12} \\ \check{I}_{21} & \check{I}_{22} \end{pmatrix},$$

where $\check{Z}_1 \in \mathbb{R}^q$, $\check{Y}_1 \in \mathbb{R}^q$, and $\check{I}_{11}$ is $q \times q$. Formula (17.17) gives

$$\check{Y}_2 = o_p(\check{Y}) = o_p(\check{Y}_1) + o_p(\check{Y}_2) \quad \text{or} \quad (1 + o_p(1)) \check{Y}_2 = o_p(\check{Y}_1),$$

which implies

$$\check{Y}_2 = o_p(\check{Y}_1).$$

Thus $o_p(\check{Y}) = o_p(\check{Y}_1)$, and (17.18) gives

$$\begin{aligned}
\check{P}\check{I}\check{Z} &= \begin{pmatrix} \check{I}_{11}\check{Z}_1 + \check{I}_{12}\check{Z}_2 \\ 0 \end{pmatrix} \\
&= \check{P}\check{I}\check{Y} + o_p(\|\check{Y}_1\|) + o_p(1) \\
&= \begin{pmatrix} \check{I}_{11}\check{Y}_1 \\ 0 \end{pmatrix} + o_p(\|\check{Y}_1\|) + o_p(1).
\end{aligned}$$

This can be written as

$$(\check{I}_{11} + o_p(1))\check{Y}_1 = \check{I}_{11}\check{Z}_1 + \check{I}_{12}\check{Z}_2 + o_p(1),$$

which implies (since $\check{I}_{11}$ is positive definite)

$$\check{Y}_1 = \check{Z}_1 + \check{I}_{11}^{-1}\check{I}_{12}\check{Z}_2 + o_p(1).$$

Note that since $\check{Z} = O_p(1)$, this equation shows that $\check{Y} = O_p(1)$, which allows us to express errors more simply in what follows. Because

$$\check{Z} - \check{Y} = \begin{pmatrix} -\check{I}_{11}^{-1}\check{I}_{12}\check{Z}_2 \\ \check{Z}_2 \end{pmatrix} + o_p(1),$$

(17.19) gives

$$\begin{aligned}
2 \log \lambda_n &= (-\check{Z}_2'\check{I}_{21}I_{11}^{-1}, \check{Z}_2') \begin{pmatrix} \check{I}_{11} & \check{I}_{12} \\ \check{I}_{21} & \check{I}_{22} \end{pmatrix} \begin{pmatrix} -\check{I}_{11}^{-1}\check{I}_{12}\check{Z}_2 \\ \check{Z}_2 \end{pmatrix} + o_p(1) \\
&= \check{Z}_2'(\check{I}_{22} - \check{I}_{21}\check{I}_{11}^{-1}\check{I}_{12})\check{Z}_2 + o_p(1).
\end{aligned}$$

Letting $\Sigma = \check{I}^{-1}$, from the formula for inverting partitioned matrices,²

$$\Sigma_{22} = (\check{I}_{22} - \check{I}_{21}\check{I}_{11}^{-1}\check{I}_{12})^{-1},$$

and so

$$2 \log \lambda_n = \check{Z}_2 \Sigma_{22}^{-1} \check{Z}_2 + o_p(1).$$

From (17.16), $\check{Z}_2 \Rightarrow N_{r-q}(\check{\Delta}_2, \Sigma_{22})$. Using Lemma 14.9,

$$2 \log \lambda_n \Rightarrow \chi_{r-q}^2(\check{\Delta}_2' \Sigma_{22}^{-1} \check{\Delta}_2).$$

This is the desired result, but for explicit computation it is convenient to express the noncentrality parameter in the original basis. Let $\check{Q} = I - \check{P}$ and $Q_0 = I - P_0$. Then

$$(\check{P} + \check{Q}\Sigma\check{Q})^{-1} = \begin{pmatrix} I_q & 0 \\ 0 & \Sigma_{22} \end{pmatrix}^{-1} = \begin{pmatrix} I_q & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix},$$

and we can express the noncentrality parameter as

² See Appendix A.6.

$$\begin{aligned}
& \Delta'_2 \Sigma_{22}^{-1} \Delta_2 \\
&= \check{\Delta}' \check{Q} \begin{pmatrix} I_q & 0 \\ 0 & \Sigma_{22}^{-1} \end{pmatrix} \check{Q} \check{\Delta} \\
&= \check{\Delta}' \check{Q} (\check{P} + \check{Q} \check{I}^{-1} \check{Q})^{-1} \check{Q} \check{\Delta} \\
&= \Delta' O O' Q_0 O (O' P_0 O + O' Q_0 O (O' I(\theta_0)^{-1} O)^{-1} O' Q_0 O)^{-1} O' Q_0 O O' \Delta \\
&= \Delta' Q_0 (P_0 + Q_0 I(\theta_0)^{-1} Q_0)^{-1} Q_0 \Delta.
\end{aligned}$$

When this formula is used in practice, it may be more convenient to substitute $I^{-1}(\theta_n)$ for $I^{-1}(\theta_0)$. Since θ_n converges to θ_0 , for large n this has negligible impact on power calculations.

17.3 Examples

Example 17.3. As a first example, suppose that $X_1, \dots, X_n, Y_1, \dots, Y_n, Z_1, \dots, Z_n$ are independent with $X_i \sim \text{Poisson}(\theta_1)$, $Y_i \sim \text{Poisson}(\theta_2)$, and $Z_i \sim \text{Poisson}(\theta_3)$, for $i = 1, \dots, n$. We can view this as a random sample of random vectors in $\mathbb{R}^3$ with density

$$f_\theta(x, y, z) = \frac{\theta_1^x \theta_2^y \theta_3^z}{x! y! z!} e^{-\theta_1 - \theta_2 - \theta_3}.$$

Then

$$\log f_\theta(x, y, z) = x \log \theta_1 + y \log \theta_2 + z \log \theta_3 - \theta_1 - \theta_2 - \theta_3 - \log(x! y! z!),$$

and

$$\nabla_\theta^2 \log f_\theta(x, y, z) = \begin{pmatrix} -x/\theta_1^2 & 0 & 0 \\ 0 & -y/\theta_2^2 & 0 \\ 0 & 0 & -z/\theta_3^2 \end{pmatrix}.$$

Hence

$$I(\theta) = -E_\theta \nabla_\theta^2 \log f_\theta(X_1, Y_1, Z_1) = \begin{pmatrix} 1/\theta_1 & 0 & 0 \\ 0 & 1/\theta_2 & 0 \\ 0 & 0 & 1/\theta_3 \end{pmatrix}.$$

Suppose we want to test $H_0 : \theta_1 + \theta_2 = \theta_3$ versus $H_1 : \theta_1 + \theta_2 \neq \theta_3$. The log-likelihood is

$$\begin{aligned}
l(\theta) &= \log(\theta_1) \sum_{i=1}^n X_i + \log(\theta_2) \sum_{i=1}^n Y_i + \log(\theta_3) \sum_{i=1}^n Z_i \\
&\quad - n\theta_1 - n\theta_2 - n\theta_3 - \sum_{i=1}^n \log(X_i! Y_i! Z_i!).
\end{aligned}$$

Maximizing l gives

$$\hat{\theta}_1 = \overline{X}, \quad \hat{\theta}_2 = \overline{Y}, \quad \hat{\theta}_3 = \overline{Z}.$$

Also, $\tilde{\theta}_1$ and $\tilde{\theta}_2$ must maximize

$$\begin{aligned} l(\theta_1, \theta_2, \theta_1 + \theta_2) &= \log(\theta_1) \sum_{i=1}^n X_i + \log(\theta_2) \sum_{i=1}^n Y_i + \log(\theta_1 + \theta_2) \sum_{i=1}^n Z_i \\ &\quad - n\theta_1 - n\theta_2 - n(\theta_1 + \theta_2) - \sum_{i=1}^n \log(X_i!Y_i!Z_i!) \end{aligned}$$

or (dividing by n and dropping the term that is independent of θ)

$$\overline{X} \log \theta_1 + \overline{Y} \log \theta_2 + \overline{Z} \log(\theta_1 + \theta_2) - 2(\theta_1 + \theta_2).$$

Setting partial derivatives with respect to θ_1 and θ_2 equal to zero gives

$$\frac{\overline{X}}{\tilde{\theta}_1} + \frac{\overline{Z}}{\tilde{\theta}_1 + \tilde{\theta}_2} = 2 \quad \text{and} \quad \frac{\overline{Y}}{\tilde{\theta}_2} + \frac{\overline{Z}}{\tilde{\theta}_1 + \tilde{\theta}_2} = 2.$$

From these equations, $\overline{X}/\tilde{\theta}_1 = \overline{Y}/\tilde{\theta}_2$. So

$$\frac{\overline{Z}}{\tilde{\theta}_1 + \tilde{\theta}_2} = \frac{\overline{Z}}{\tilde{\theta}_2(1 + \overline{X}/\overline{Y})}.$$

Using this in the second normal equation,

$$2\tilde{\theta}_2 = \overline{Y} + \frac{\overline{Z}}{1 + \overline{X}/\overline{Y}} = \overline{Y} \left(\frac{\overline{Z} + \overline{X} + \overline{Y}}{\overline{X} + \overline{Y}} \right).$$

Hence

$$\tilde{\theta}_1 = \frac{\overline{X}}{2} \left(\frac{\overline{Z} + \overline{X} + \overline{Y}}{\overline{X} + \overline{Y}} \right), \quad \tilde{\theta}_2 = \frac{\overline{Y}}{2} \left(\frac{\overline{Z} + \overline{X} + \overline{Y}}{\overline{X} + \overline{Y}} \right),$$

and

$$\tilde{\theta}_3 = \frac{\overline{Z} + \overline{X} + \overline{Y}}{2}.$$

Now

$$\begin{aligned} 2 \log \lambda &= 2(l(\hat{\theta}) - l(\tilde{\theta})) \\ &= 2n \left[\overline{X} \log(\hat{\theta}_1/\tilde{\theta}_1) + \overline{Y} \log(\hat{\theta}_2/\tilde{\theta}_2) + \overline{Z} \log(\hat{\theta}_3/\tilde{\theta}_3) \right. \\ &\quad \left. + \tilde{\theta}_1 + \tilde{\theta}_2 + \tilde{\theta}_3 - \hat{\theta}_1 - \hat{\theta}_2 - \hat{\theta}_3 \right]. \end{aligned}$$

Since $\tilde{\theta}_1 + \tilde{\theta}_2 + \tilde{\theta}_3 = \hat{\theta}_1 + \hat{\theta}_2 + \hat{\theta}_3$, this simplifies to

$$2 \log \lambda = -2n \left[(\overline{X} + \overline{Y}) \log \left(\frac{\overline{Z} + \overline{X} + \overline{Y}}{2\overline{X} + 2\overline{Y}} \right) + \overline{Z} \log \left(\frac{\overline{Z} + \overline{X} + \overline{Y}}{2\overline{Z}} \right) \right].$$

In this example, $r = 3$ and $q = 2$, so under H_0 , $2 \log \lambda$ is approximately χ_1^2 . If c is the $1 - \alpha$ quantile of χ_1^2 , then the test that rejects if $2 \log \lambda > c$ has size approximately α . To approximate the power of this test using the results from the last section we need to identify the projection matrices that arise. Because Ω_0 is linear, the tangent space $V = V_{\theta_0}$ is the same for all $\theta_0 \in \Omega_0$. The vectors

$$v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad \text{and} \quad v_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

form an orthonormal basis for $\mathbb{R}^3$. Both v_1 and v_2 lie in the tangent space V , and v_3 lies in $V^\perp$. So

$$P_0 = v_1 v_1' + v_2 v_2' = \frac{1}{3} \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

and

$$Q_0 = v_3 v_3' = \frac{1}{3} \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}.$$

Suppose $\theta = \theta_0 + \Delta/\sqrt{n}$, where $\theta_0 \in \Omega_0$. Then

$$Q_0 \theta = Q_0 \theta_0 + Q_0 \Delta/\sqrt{n},$$

which implies

$$Q_0 \Delta = \sqrt{n} Q_0 \theta = \sqrt{n} v_3 (v_3' \theta).$$

Since

$$I^{-1}(\theta) = \begin{pmatrix} \theta_1 & 0 & 0 \\ 0 & \theta_2 & 0 \\ 0 & 0 & \theta_3 \end{pmatrix},$$

we have

$$\begin{aligned} Q_0 I^{-1}(\theta) Q_0 &= v_3 (v_3' I^{-1}(\theta) v_3) v_3' \\ &= \frac{\theta_1 + \theta_2 + \theta_3}{3} v_3 v_3' \\ &= \frac{\theta_1 + \theta_2 + \theta_3}{3} Q_0. \end{aligned}$$

Hence

$$\begin{aligned} (P_0 + Q_0 I^{-1}(\theta) Q_0)^{-1} &= \left(P_0 + \frac{\theta_1 + \theta_2 + \theta_3}{3} Q_0 \right)^{-1} \\ &= P_0 + \frac{3}{\theta_1 + \theta_2 + \theta_3} Q_0. \end{aligned}$$

The formula for the noncentrality parameter (substituting $I^{-1}(\theta)$ for $I^{-1}(\theta_0)$) is

$$\begin{aligned}\Delta' Q_0 \left(P_0 + \frac{3}{\theta_1 + \theta_2 + \theta_3} Q_0 \right) Q_0 \Delta &= \frac{3 \Delta' Q_0 Q_0 \Delta}{\theta_1 + \theta_2 + \theta_3} \\ &= \frac{3 \|Q_0 \Delta\|^2}{\theta_1 + \theta_2 + \theta_3} \\ &= \frac{3n(v'_3 \theta)^2}{\theta_1 + \theta_2 + \theta_3} \\ &= \frac{n(\theta_1 + \theta_2 - \theta_3)^2}{\theta_1 + \theta_2 + \theta_3}.\end{aligned}$$

To be concrete, suppose a test with size 5% is desired. Then one would take $c = 1.96^2 = 3.84$. If $\theta_1 = \theta_2 = 1$, $\theta_3 = 2.3$, and $n = 100$, then the noncentrality parameter comes out $\delta^2 = 9/4.3 = 2.09 = 1.45^2$. If we let $Z \sim N(0, 1)$, then $(Z + 1.45)^2 \sim \chi_1^2(2.09)$ and the power of the test is approximately

$$\begin{aligned}P(2 \log \lambda > 3.84) &\approx P\{(Z + 1.45)^2 > 1.96^2\} \\ &= P(Z > .51) + P(Z < -3.41) = 0.3053.\end{aligned}$$

Example 17.4. Our final example concerns the classic problem of testing independence in two-way contingency tables. The data are

$$\begin{pmatrix} N_{11} \\ N_{12} \\ N_{21} \\ N_{22} \end{pmatrix} \sim \text{Multinomial}(n; p_{11}, p_{12}, p_{21}, p_{22}).$$

Because the p_{ij} must sum to one, they lie on the unit simplex in $\mathbb{R}^4$. This set is not open, so to apply our results directly we take

$$\theta = \begin{pmatrix} p_{11} \\ p_{12} \\ p_{21} \end{pmatrix}.$$

The maximum likelihood estimates for the p_{ij} are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}.$$

Here we follow the common convention where a “+” as a subscript indicates that terms should be summed; so $p_{i+} = p_{i1} + p_{i2}$ and $N_{+j} = N_{1j} + N_{2j}$, for example. The null hypothesis of independence in the table is

$$H_0 : p_{ij} = p_{i+} p_{+j}, \text{ for } i = 1, 2 \text{ and } j = 1, 2.$$

Equivalently,

$$H_0 : p_{11} = p_{1+}p_{+1}. \quad (17.20)$$

(For instance, if (17.20) holds, then $p_{12} = p_{1+} - p_{11} = p_{1+} - p_{1+}p_{+1} = p_{1+}(1 - p_{+1}) = p_{1+}p_{+2}$.) The log-likelihood function is

$$l = \sum_{i,j} N_{ij} \log p_{ij} + \log \binom{n}{N_{11}, \dots, N_{22}}.$$

Under H_0 ,

$$\begin{aligned} l &= \sum_{ij} N_{ij} \log(p_{i+}p_{+j}) + \log \binom{n}{N_{11}, \dots, N_{22}} \\ &= \sum_{ij} N_{ij} \log(p_{i+}) + \sum_{ij} N_{ij} \log(p_{+j}) + \log \binom{n}{N_{11}, \dots, N_{22}} \\ &= \sum_i N_{i+} \log(p_{i+}) + \sum_j N_{+j} \log(p_{+j}) + \log \binom{n}{N_{11}, \dots, N_{22}} \\ &= N_{1+} \log(p_{1+}) + N_{2+} \log(1 - p_{1+}) + N_{+1} \log(p_{+1}) \\ &\quad + N_{+2} \log(1 - p_{+1}) + \log \binom{n}{N_{11}, \dots, N_{22}}. \end{aligned}$$

Setting partial derivatives with respect to p_{+1} and p_{1+} to zero gives the following normal equations for $\tilde{p}_{1+}$ and $\tilde{p}_{+1}$:

$$\frac{N_{1+}}{\tilde{p}_{1+}} - \frac{N_{2+}}{1 - \tilde{p}_{1+}} = 0$$

and

$$\frac{N_{1+}}{\tilde{p}_{+1}} - \frac{N_{+2}}{1 - \tilde{p}_{+1}} = 0.$$

Solving these equations,

$$\tilde{p}_{1+} = \frac{N_{1+}}{n} = \hat{p}_{1+}$$

and

$$\tilde{p}_{+1} = \frac{N_{+1}}{n} = \hat{p}_{+1}.$$

It follows that $\tilde{p}_{+j} = \hat{p}_{+j}$ for $j = 1, 2$ and $\tilde{p}_{i+} = \hat{p}_{i+}$ for $i = 1, 2$. Therefore

$$\tilde{p}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$$

for $i = 1, 2$ and $j = 1, 2$. Plugging in the maximum likelihood estimates derived gives

$$\begin{aligned}
2 \log \lambda &= 2l(\hat{\theta}) - 2l(\tilde{\theta}) \\
&= 2 \sum_{i,j} N_{ij} \log(\hat{p}_{ij}) - 2 \sum_{i,j} N_{ij} \log(\hat{p}_{i+} \hat{p}_{+j}) \\
&= 2 \sum_{i,j} N_{ij} \log\left(\frac{\hat{p}_{ij}}{\hat{p}_{i+} \hat{p}_{+j}}\right).
\end{aligned}$$

Let us now turn our attention to the approximate distribution of the likelihood ratio test statistic. Because Ω_0 has dimension $q = 2$ and Ω is an open set in $\mathbb{R}^3$, under H_0 ,

$$2 \log \lambda \sim \chi_1^2.$$

To approximate the power at contiguous alternatives we need the Fisher information. We know that

$$\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N_3(0, I^{-1}(\theta))$$

as $n \rightarrow \infty$. Since $\hat{\theta}$ can be viewed as an average of n i.i.d. vectors, by the central limit theorem,

$$\sqrt{n}(\hat{\theta} - \theta) \Rightarrow N_3(0, \Sigma)$$

as $n \rightarrow \infty$, where Σ is the covariance of $\hat{\theta}$ when $n = 1$. With $n = 1$,

$$\begin{aligned}
\text{Cov}(\hat{p}_{ij}, \hat{p}_{kl}) &= \text{Cov}(N_{ij}, N_{kl}) \\
&= EN_{ij}N_{kl} - p_{ij}p_{kl} \\
&= \begin{cases} -p_{ij}p_{kl}, & (i, j) \neq (k, l); \\ p_{ij}(1 - p_{ij}), & i = k, j = l. \end{cases}
\end{aligned}$$

Letting $q_{ij} = 1 - p_{ij}$, we have

$$\Sigma = I^{-1}(\theta) = \begin{pmatrix} p_{11}q_{11} & -p_{11}p_{12} & -p_{11}p_{21} \\ -p_{11}p_{12} & p_{12}q_{12} & -p_{12}p_{21} \\ -p_{11}p_{21} & -p_{12}p_{21} & p_{21}q_{21} \end{pmatrix}.$$

Fix $\theta_0 \in \Omega_0$ and let V_0 be the tangent space for Ω_0 at θ_0 . To identify the projection matrices P_0 and Q_0 onto V_0 and $V_0^\perp$, note that parameters $\theta \in \Omega_0$ must satisfy the constraint

$$\theta_1 - (\theta_1 + \theta_2)(\theta_1 + \theta_3) \stackrel{\text{def}}{=} g(\theta) = 0.$$

Using results from Appendix A.4, $V_0^\perp$ is the space spanned by the rows of $Dg(\theta_0)$, or the columns of $Dg(\theta_0)' = \nabla g(\theta_0)$. Direct calculation gives

$$\nabla g(\theta) = \begin{pmatrix} 1 - \theta_2 - \theta_3 - 2\theta_1 \\ -(\theta_1 + \theta_3) \\ -(\theta_1 + \theta_2) \end{pmatrix}.$$

Let

$$v_3 = \frac{\nabla g(\theta_0)}{\|\nabla g(\theta_0)\|},$$

and choose v_1 and v_2 so that $\{v_1, v_2, v_3\}$ is an orthonormal basis for $\mathbb{R}^3$. As in Example 17.3, v_1 and v_2 span V_0 , v_3 spans $V_0^\perp$,

$$P_0 = v_1 v_1' + v_2 v_2'$$

and

$$Q_0 = v_3 v_3'.$$

The noncentrality parameter is

$$\begin{aligned} \delta^2 &= \Delta' Q_0 (P_0 + Q_0 I^{-1}(\theta_0) Q_0)^{-1} Q_0 \Delta \\ &= \Delta' v_3 v_3' (P_0 + v_3 v_3' I^{-1}(\theta_0) v_3 v_3')^{-1} v_3 v_3' \Delta \\ &= (v_3' \Delta)^2 v_3' (P_0 + [v_3' I^{-1}(\theta_0) v_3] Q_0)^{-1} v_3 \\ &= (v_3' \Delta)^2 v_3' \left(P_0 + \frac{Q_0}{v_3' I^{-1}(\theta_0) v_3} \right) v_3 \\ &= \frac{(v_3' \Delta)^2}{v_3' I^{-1}(\theta_0) v_3} \\ &= \frac{n(\nabla g(\theta_0) \cdot (\theta_n - \theta_0))^2}{\nabla g(\theta_0)' I^{-1}(\theta_0) \nabla g(\theta_0)}. \end{aligned}$$

The derivation leading to this formula works whenever $r - q = 1$ with Ω_0 the parameters $\theta \in \Omega$ satisfying a single differentiable constraint $g(\theta) = 0$.

To illustrate use of the distributional results in a more concrete setting, let us consider the following design question. How large should the sample size be to achieve a test with (approximate) level $\alpha = 5\%$ and power 90% when $p_{11} = p_{22} = 0.3$ and $p_{12} = p_{21} = 0.2$? The parameter value associated with these cell probabilities is

$$\theta_n = \begin{pmatrix} 0.3 \\ 0.2 \\ 0.2 \end{pmatrix}.$$

Under H_0 , $2 \log \lambda \sim Z^2$, where $Z \sim N(0, 1)$. Since $P(Z^2 > (1.96)^2) = 1 - P(-1.96 < Z < 1.96) = 5\% = \alpha$, the test should reject if $2 \log \lambda > (1.96)^2$. Since $(Z + \delta)^2 \sim \chi_1^2(\delta^2)$, under the alternative θ_n ,

$$2 \log \lambda \sim (Z + \delta)^2.$$

To meet the design objective, we need

$$\begin{aligned} 90\% &\approx P((Z + \delta)^2 > (1.96)^2) \\ &= P(Z > 1.96 - \delta) + P(Z < -1.96 - \delta). \end{aligned}$$

The second term here is negligible, so we require

$$90\% \approx P(Z > 1.96 - \delta).$$

This holds if $1.96 - \delta$ is the 10th percentile of the standard normal distribution. This percentile is -1.282 which gives $\delta = 1.96 + 1.282 = 3.242$ and $\delta^2 = 10.51$. The marginal cell probabilities under θ_n are $p_{i+} = p_{+j} = 1/2$, so the natural choice for θ_0 is

$$\theta_0 = \begin{pmatrix} 1/4 \\ 1/4 \\ 1/4 \end{pmatrix}.$$

Then

$$\nabla g(\theta_0) = \begin{pmatrix} 0 \\ -1/2 \\ -1/2 \end{pmatrix}$$

and

$$\nabla g(\theta_0) \cdot (\theta_n - \theta_0) = \begin{pmatrix} 0 \\ -1/2 \\ -1/2 \end{pmatrix} \cdot \begin{pmatrix} 1/20 \\ -1/20 \\ -1/20 \end{pmatrix} = \frac{1}{20}.$$

Also,

$$I^{-1}(\theta_0) = \frac{1}{4} \begin{pmatrix} 3/4 & -1/4 & -1/4 \\ -1/4 & 3/4 & -1/4 \\ -1/4 & -1/4 & 3/4 \end{pmatrix},$$

and so

$$\nabla g(\theta_0)' I^{-1}(\theta_0) \nabla g(\theta_0) = \frac{1}{4} \left(\frac{3}{4} - \frac{1}{4} - \frac{1}{4} + \frac{3}{4} \right) = \frac{1}{16}.$$

Hence

$$\delta^2 = \frac{n(1/20)^2}{(1/4)^2} = \frac{n}{25}.$$

Setting this equal to 10.51 gives $n = 263$ as the sample size required for the level and power specified.

In practice many statisticians test independence in 2×2 tables using Pearson's chi-square test statistic,

$$T = \sum_{i,j} \frac{(N_{ij} - n\hat{p}_{i+}\hat{p}_{+j})^2}{n\hat{p}_{i+}\hat{p}_{+j}}.$$

For large n , T and $2 \log \lambda$ are asymptotically equivalent. To demonstrate this equivalence (without any serious attempt at mathematical rigor), we write

$$2 \log \lambda = 2n \sum_{i,j} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right).$$

We view this as a function of the $\hat{p}_{ij}$ with the marginal probabilities $\hat{p}_{i+}$ and $\hat{p}_{+j}$ considered *fixed constants* and Taylor expand about $\hat{p}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$ (equality

here is approximately correct under both H_0 and contiguous alternatives). To compute the gradient of the function,

$$\frac{\partial}{\partial \hat{p}_{kl}}(2 \log \lambda) = 2n \left[\log \left(\frac{\hat{p}_{kl}}{\hat{p}_{k+} \hat{p}_{+l}} \right) + 1 \right]. \quad (17.21)$$

Then

$$\left. \frac{\partial}{\partial \hat{p}_{ij}}(2 \log \lambda) \right|_{\hat{p}_{ij} = \hat{p}_{i+} \hat{p}_{+j}} = 2n,$$

and the gradient at the point of expansion is

$$\begin{pmatrix} 2n \\ 2n \\ 2n \\ 2n \end{pmatrix}.$$

Taylor expansion through the gradient term gives

$$2 \log \lambda \approx 2n \sum_{i,j} (\hat{p}_{ij} - \hat{p}_{i+} \hat{p}_{+j}) = 0.$$

To get an interesting answer we need to keep an extra term in our Taylor expansion. Because (17.21) only depends on $\hat{p}_{ij}$, the Hessian matrix is diagonal. Now

$$\frac{\partial^2}{\partial \hat{p}_{ij}^2}(2 \log \lambda) = \frac{2n}{\hat{p}_{ij}},$$

so

$$\left. \frac{\partial^2}{\partial \hat{p}_{ij}^2}(2 \log \lambda) \right|_{\hat{p}_{ij} = \hat{p}_{i+} \hat{p}_{+j}} = \frac{2n}{\hat{p}_{i+} \hat{p}_{+j}}.$$

Taylor expansion through the Hessian term gives

$$2 \log \lambda \approx \frac{1}{2} \sum_{i,j} \frac{2n}{\hat{p}_{i+} \hat{p}_{+j}} (\hat{p}_{ij} - \hat{p}_{i+} \hat{p}_{+j})^2 = T.$$

17.4 Wald and Score Tests

The Wald and score (or Lagrange multiplier) tests are alternatives to the generalized likelihood ratio tests with similar properties. Assume that the null hypothesis can be written as $H_0 : g(\theta) = 0$, with the constraint function $g : \Omega \rightarrow \mathbb{R}^{r-q}$ continuously differentiable and $Dg(\theta)$ of full rank for $\theta \in \Omega_0$.

The basic idea behind the Wald test (Wald (1943)) is simply that if H_0 is correct, then $g(\hat{\theta}_n)$ should be close to zero. By Proposition 9.32, if $\theta \in \Omega_0$, then

$$\sqrt{n}g(\hat{\theta}_n) \Rightarrow N \left[0, Dg(\theta)I(\theta)^{-1}(Dg(\theta))' \right].$$

By Lemma 14.9 and results on weak convergence,

$$T_W \stackrel{\text{def}}{=} n(g(\hat{\theta}_n))' \left[Dg(\hat{\theta}_n)I(\hat{\theta}_n)^{-1}(Dg(\hat{\theta}_n))' \right]^{-1} g(\hat{\theta}_n) \Rightarrow \chi_{r-q}^2$$

when $\theta \in \Omega_0$.

Rao's score test (Rao (1948)) is based on the notion that if $\theta \in \Omega_0$, then $\tilde{\theta}_n$ should be a good estimate of θ and the gradient of the log-likelihood should not be too large at $\tilde{\theta}_n$. Differencing (17.12) and (17.11), for $\theta \in \Omega_0$

$$\frac{1}{\sqrt{n}}\nabla l_n(\tilde{\theta}_n) = I(\theta)(Z_n - Y_n) + o_p(1)$$

as $n \rightarrow \infty$. Using (17.13) it is then not hard to show that

$$T_S \stackrel{\text{def}}{=} \frac{1}{n}(\nabla l_n(\tilde{\theta}_n))' I(\tilde{\theta}_n)^{-1} \nabla l_n(\tilde{\theta}_n) = 2 \log \lambda_n + o_p(1) \Rightarrow \chi_{r-q}^2.$$

The three test statistics, $2 \log \lambda$, T_W , and T_S , have different strengths and weaknesses. Although the derivation here only considers the asymptotic null distributions of T_W and T_S , with the methods and regularity assumed in Section 17.2 it is not hard to argue that all three tests are asymptotically equivalent under distributions contiguous to a null distribution; specifically, differences between any two of the statistics will tend to zero in probability. Furthermore, variants of T_W and T_S in which the Fisher information is estimated consistently in a different fashion, perhaps using observed Fisher information, are also equivalent under distributions contiguous to a null distribution.

The score test only relies on $\tilde{\theta}_n$. The maximum likelihood estimator $\hat{\theta}_n$ under the full model is not needed. This may be advantageous if the full model is difficult to fit. Unfortunately, it also means that although the test will have good power at alternatives near a null distribution, the power may not be high at more distant alternatives. In fact, there are examples where the power of the score test does not tend to one at fixed alternatives as $n \rightarrow \infty$. See Freedman (2007).

In contrast to the score test, the Wald test statistic relies only on the maximum likelihood estimator under the full model $\hat{\theta}_n$, and there is no need to compute $\tilde{\theta}_n$. This may make T_W easier to compute than $2 \log \lambda$.

With a fixed nominal level α , at a fixed alternative $\theta \in \Omega_1$ the powers of the generalized likelihood, Wald, and the score test generally tend to one quickly. With sufficient regularity, the convergence occurs exponentially quickly, with the generalized likelihood ratio test having the best possible rate of convergence. This rate of convergence is called the *Bahadur slope* for the test, and the generalized likelihood ratio is thus *Bahadur efficient*. But from a practical standpoint, the ability of a test to detect smaller differences may be more important, and in this regard it is harder to say which of these tests is best.

17.5 Problems³

1. Consider three samples: $W_1, \dots, W_k$ from $N(\mu_1, \sigma_1^2)$; $X_1, \dots, X_m$ from $N(\mu_1, \sigma_2^2)$; and $Y_1, \dots, Y_n$ from $N(\mu_2, \sigma_2^2)$, all independent, where μ_1, μ_2, σ_1 , and σ_2 are unknown parameters. Derive the generalized likelihood test statistic λ to test $H_0 : \sigma_1 = \sigma_2$ versus $H_1 : \sigma_1 \neq \sigma_2$. You should be able to reduce the normal equations under the full model to a single cubic equation. Explicit solution of this cubic equation is not necessary.
2. Consider data for a two-way contingency table $N_{11}, N_{12}, N_{21}, N_{22}$ from a multinomial distribution with n trials and success probabilities $p_{11}, p_{12}, p_{21}, p_{22}$. Derive the generalized likelihood test statistic λ to test “symmetry,” $H_0 : p_{12} = p_{21}$ versus $H_1 : p_{12} \neq p_{21}$.
- *3. *Random effects models.* One model that is used to analyze a blocked experiment comparing p treatments has $Y_{ij} = \alpha_i + \beta_j + \epsilon_{ij}$, $i = 1, \dots, p$, $j = 1, \dots, n$, with the α_i and β_j viewed as unknown constant parameters and the ϵ_{ij} unobserved and i.i.d. from $N(0, \sigma^2)$. In some circumstances, it may be more natural to view the blocking variables β_j as random, perhaps as i.i.d. from $N(0, \tau^2)$ and independent of the ϵ_{ij} . This gives a model in which the vectors $Y_j = (Y_{1j}, \dots, Y_{pj})'$, $j = 1, \dots, n$, are i.i.d. from $N(\alpha, \sigma^2 I + \tau^2 \mathbf{1}\mathbf{1}')$. Here “ $\mathbf{1}$ ” denotes a column of 1s in $\mathbb{R}^p$, and the unknown parameters are $\alpha \in \mathbb{R}^p$, $\sigma^2 > 0$, and $\tau^2 \geq 0$.
 - a) Derive the likelihood ratio test statistic to test $H_0 : \tau^2 = 0$ versus $H_1 : \tau^2 > 0$.
 - b) Derive the likelihood ratio test statistic to test $H_0 : \alpha_1 = \dots = \alpha_p$.
- *4. Let X and Y be independent exponential variables with failure rates θ_x and θ_y , respectively.
 - a) Find the generalized likelihood ratio test statistic λ , based on X and Y , to test $H_0 : \theta_x = 2\theta_y$ versus $H_1 : \theta_x \neq 2\theta_y$.
 - b) Suppose the test rejects if $\lambda \geq c$. How should the critical level c be adjusted to give level α ?
5. Let X and Y be independent normal variables, both with variance one and means $\theta_X = EX$ and $\theta_Y = EY$.
 - a) Derive the generalized likelihood test statistic λ (or $\log \lambda$) to test $H_0 : \theta_X = 0$ or $\theta_Y = 0$ versus $H_1 : \theta_X \neq 0$ and $\theta_Y \neq 0$.
 - b) The likelihood test using λ from part (a) rejects H_0 if $\log \lambda > k$. Derive a formula for the power of this test when $\theta_X = 0$.
 - c) Find the significance level α as a function of k . How should k be chosen to achieve a desired level α ?
6. Suppose $X \sim N_p(\theta, I)$ and consider testing $H_0 : \theta \in \Omega_0$ versus $H_1 : \theta \notin \Omega_0$.
 - a) Show that the likelihood ratio test statistic λ is equivalent to the distance D between X and Ω_0 , defined as

$$D = \inf \{ \|X - \theta\| : \theta \in \Omega_0 \}.$$

³ Solutions to the starred problems are given at the back of the book.

(Equivalent here means there is a one-to-one increasing relationship between the two statistics.)

- b) Using part (a), a generalized likelihood ratio test will reject H_0 if $D > c$. What is the significance level α for this test if $p = 2$ and $\Omega_0 = \{\theta : \theta_1 \leq 0, \theta_2 \leq 0\}$?
7. Show that in the general linear model there is an increasing one-to-one relationship between the generalized likelihood ratio statistic λ and the test statistic T in (14.24), so that tests based on λ and T are equivalent.
- *8. Let $X_1, \dots, X_n$ be a random sample from an exponential distribution with mean θ_1 , and let $Y_1, \dots, Y_n$ be an independent sample from an exponential distribution with mean θ_2 .
 - a) Find the likelihood ratio test statistic for testing $H_0 : \theta_1/\theta_2 = c_0$ versus $H_1 : \theta_1/\theta_2 \neq c_0$, where c_0 is a constant.
 - b) Use the large-sample approximation for the null distribution of $2 \log \lambda$ and the duality between testing and interval estimation to describe a confidence set for θ_1/θ_2 with coverage probability approximately 95% (the set is an interval, but you do not have to demonstrate this fact). If $n = 100$, $\bar{X} = 2$, and $\bar{Y} = 1$, determine whether the parameter ratio 2.4 lies in the confidence set.
 - c) How large should the sample size n be if we want the likelihood ratio test for testing $H_0 : \theta_1 = \theta_2$ versus $H_1 : \theta_1 \neq \theta_2$ at level 5% to have power 90% when $\theta_1 = 0.9$ and $\theta_2 = 1.1$?
- *9. Let $W_1, \dots, W_n, X_1, \dots, X_n$, and $Y_1, \dots, Y_n$ be independent random samples from $N(\mu_w, \sigma_w^2)$, $N(\mu_x, \sigma_x^2)$, and $N(\mu_y, \sigma_y^2)$, respectively.
 - a) Find the likelihood ratio test statistic for testing $H_0 : \sigma_w = \sigma_x = \sigma_y$ versus the alternative that at least two of the standard deviations differ.
 - b) What is the approximate power of the likelihood ratio test with level $\alpha = 5\%$ if $n = 200$, $\sigma_w = 1.8$, $\sigma_x = 2.2$ and $\sigma_y = 2.0$? You can express the answer in terms of a noncentral chi-square distribution, but identify the appropriate degrees of freedom and the noncentrality parameter.
- *10. Suppose $X_1, \dots, X_n$ are i.i.d. $N_p(\mu, I)$.
 - a) Derive the likelihood ratio test statistic $2 \log \lambda$ to test $H_0 : \|\mu\| = r$ versus $H_1 : \|\mu\| \neq r$, where r is a fixed constant.
 - b) Give a formula for the power of the likelihood ratio test that rejects H_0 when $2 \log \lambda > c$ in terms of the cumulative distribution function for a noncentral chi-square distribution.
 - c) If $r = 1$, what sample size will be necessary for the test with $\alpha \approx 5\%$ to have power approximately 90% when $\|\mu\| = 1.1$?
11. Let $W_1, \dots, W_n, X_1, \dots, X_n$, and $Y_1, \dots, Y_n$ be independent random samples. The W_i have density $e^{-|x-\theta_1|}/2$, the Y_i have density $e^{-|x-\theta_2|}/2$, and the X_i have density $e^{-|x-\theta_3|}/2$. Derive the approximate power for the likelihood ratio test with $\alpha = 5\%$ of $H_0 : \theta_1 = \theta_2 = \theta_3$ if $n = 200$, $\theta_1 = 1.8$, $\theta_2 = 2.0$, and $\theta_3 = 2.2$. You can express the answer in terms of

a noncentral chi-square distribution, but identify the appropriate degrees of freedom and the noncentrality parameter.

- *12. *Errors in variables models.* Consider a regression model in which

$$Y_i = \beta X_i + \epsilon_i, \quad i = 1, \dots, n,$$

with the X_i a random sample from $N(0, 1)$ and the ϵ_i an independent random sample, also from $N(0, 1)$. In some situations, the independent variables X_i may not be observed directly. One possibility is that they are measured with error. For a specific model, let

$$W_i = X_i + \eta_i, \quad i = 1, \dots, n.$$

The η_i are modeled as a random sample from $N(0, \sigma^2)$, independent of the X_i and ϵ_i . The data are $W_1, \dots, W_n$ and $Y_1, \dots, Y_n$, with β and σ unknown parameters.

- Determine the joint distribution of W_i and Y_i .
 - Describe how to compute the generalized likelihood ratio test statistic to test $H_0 : \beta = 0$ versus $H_1 : \beta \neq 0$. An explicit formula for the maximum likelihood estimators may not be feasible, but you should give equations that can be solved to find the maximum likelihood estimators.
 - Show that the least squares estimate for β when $\sigma = 0$ (the estimator that one would use when the model ignores measurement error for the independent variable) is inconsistent if $\sigma > 0$.
 - Derive an approximation for the power of the generalized likelihood ratio test with level $\alpha \approx 5\%$ when $\beta = \Delta/\sqrt{n}$. How does σ effect the power of the test?
- *13. *Goodness-of-fit test.* Let $X_1, \dots, X_n$ be a random sample from some continuous distribution on $(0, \infty)$, and let Y_1 be the number of observations in $(0, 1)$, Y_2 the number of observations in $[1, 2)$, and Y_3 the number of observations in $[2, \infty)$. Then Y has a multinomial distribution with n trials and success probabilities p_1, p_2, p_3 . If the distribution of the X_i is exponential with failure rate θ , then $p_1 = 1 - e^{-\theta}$, $p_2 = e^{-\theta} - e^{-2\theta}$, and $p_3 = e^{-2\theta}$.
- Derive a generalized likelihood ratio test of the null hypothesis that the X_i come from an exponential distribution. The test should be based on data Y_1, Y_2, Y_3 .
 - If $\alpha \approx 5\%$ and $Y = (36, 24, 40)$, would the test in part (a) accept or reject the null hypothesis? What is the attained significance with these data (approximately)?
 - How large should the sample size be if we want the test with level $\alpha \approx 5\%$ to have power 90% when $p_1 = 0.36$, $p_2 = 0.24$, and $p_3 = 0.4$?
14. Let $X_1, \dots, X_n$ be i.i.d. from a Fisher distribution on the unit sphere in $\mathbb{R}^3$ with common density

$$f_\theta(x) = \frac{\|\theta\| e^{\theta \cdot x}}{4\pi \sinh(\|\theta\|)},$$

with respect to surface area on the unit sphere. When $\theta = 0$, $f_0(x) = 1/(4\pi)$, so the variables are uniformly distributed. (These distributions are often used to model solid angles.)

- a) Describe how you would test $H_0 : \theta_2 = \theta_3 = 0$ versus $H_1 : \theta_2 \neq 0$ or $\theta_3 \neq 0$, giving the normal equations you would use to solve for $\hat{\theta}_n$ and $\hat{\theta}_n$. If $\alpha = 5\%$, $n = 100$, and $\bar{X}_n = (0.6, 0.1, 0.1)$, would you accept or reject H_0 ?
 - b) What is the approximate power of the likelihood test if $n = 100$, $\alpha = 5\%$, and $\theta = (1, .2, 0)$? Express the answer using the noncentral chi-square distribution, but identify the degrees of freedom and the noncentrality parameter.
15. Let X_{ij} , $i = 1, \dots, p$, $j = 1, \dots, n$, be i.i.d. from a standard exponential distribution, and given $X = x$, let Y_{ij} , $i = 1, \dots, p$, $j = 1, \dots, n$, be independent Poisson variables with $EY_{ij} = x_{ij}\theta_i$. Then $(X_{1j}, \dots, X_{pj}, Y_{1j}, \dots, Y_{pj})$, $j = 1, \dots, n$, are i.i.d. random vectors. Consider testing $H_0 : \theta_1 = \dots = \theta_p$ versus $H_1 : \theta_i \neq \theta_k$ for some $i \neq k$.
- a) Find the likelihood ratio test statistic to test H_0 against H_1 .
 - b) What is the approximate power for the likelihood ratio test if $\alpha = 5\%$, $n = 100$, $p = 5$, $\theta_1 = 1.8$, $\theta_2 = 1.9$, $\theta_3 = 2.0$, $\theta_4 = 2.1$, and $\theta_5 = 2.2$? Express your answer using the noncentral chi-square distribution, but give the degrees of freedom and the noncentrality parameter.
 - c) If $p = 2$ and $\alpha \approx 5\%$, how large should the sample size n be if power 90% is desired when $\theta_1 = 1.9$ and $\theta_2 = 2.1$?
16. Define

$$\Omega_0 = \{x \in (0, \infty)^3 : x_1 x_2 x_3 = 10\},$$

a manifold in $\mathbb{R}^3$.

- a) What is the dimension of Ω_0 ?
 - b) Let V be the tangent space for this manifold at $x = (1, 2, 5)$. Find an orthonormal basis e_1, e_2, e_3 with $e_1, \dots, e_q$ spanning V and $e_{q+1}, \dots, e_3$ spanning $V^\perp$.
 - c) Find projection matrices P and Q onto V and $V^\perp$, respectively.
17. Define a vector-valued function $\eta : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $\eta_1(x, y) = x^2$, $\eta_2(x, y) = y^2$, and $\eta_3(x, y) = (x - y)^2$, and let $\Omega_0 = \eta((0, \infty)^2)$. Let V be the tangent space for Ω_0 at $(4, 1, 1)$. Find orthonormal vectors that span V .
18. Let $(N_{11}, \dots, N_{22})$ have a multinomial distribution, as in Example 17.4, and consider testing whether the marginal distributions are the same, that is, testing $H_0 : p_{+1} = p_{1+}$ versus $H_1 : p_{+1} \neq p_{1+}$.
- a) Derive a formula for the generalized likelihood ratio test statistic $2 \log \lambda$.
 - b) How large should the number of trials be if we want the test with level $\alpha \approx 5\%$ to have power 90% when $p_{11} = 30\%$, $p_{12} = 15\%$, $p_{21} = 20\%$, and $p_{22} = 35\%$?